Dinh Trong Nghia

Principal Speech Scientist

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SUMMARY OF QUALIFICATIONS

PhD in signal processing with 9 years in speech and audio machine learning, including 4 years leading a research team. I have taken wake word, speaker verification and diarization systems from research to production on real devices.

AREAS OF COMPETENCE

Speech: KWS, VAD, ECAPA-TDNN, x-vectors, WavLM, anti-spoofing, pyannote, NVIDIA NeMo, Sortformer, EEND
Real-time voice: Streaming ASR-LLM-TTS, full-duplex speech LLMs, barge-in, AEC, beamforming
Engineering: Python, PyTorch, C++, ONNX, NVIDIA Jetson
Evaluation: FRR @ FA/hour, EER, minDCF, DER/JER, WER/CER, ROC/DET analysis

CAREER HISTORY

Principal Speech Scientist Jan 2022 – Present

Orion Robotics Lab, Hanoi

- Lead a team of 8 and own the speech stack for service robots, from audio front-end to voice interaction.
- Shipped a bilingual Vietnamese-English wake word model on Jetson at under 0.5 false alarms per hour.
- Integrated a full-duplex speech LLM and a streaming ASR-LLM-TTS pipeline, then compared latency and barge-in.
- Built Vietnamese speech datasets covering regional accents, code-switching and far-field noise.

Senior Speech Research Engineer Sep 2017 – Dec 2021

Sonus Devices Asia, Singapore

- Delivered speaker verification with ASVspoof-style anti-spoofing on consumer devices.
- Wrote real-time C++ inference for overlap-aware streaming diarization.

SELECTED PUBLICATIONS AND PATENTS

- Papers at Interspeech and ICASSP on streaming diarization and low-power keyword spotting.
- Two granted patents on barge-in detection for robots.

EDUCATION

PhD in Electrical Engineering (Speech Signal Processing), Nanyang Technological University, 2013 – 2017
B.Eng. in Electronics and Telecommunications, Hanoi University of Science and Technology, 2008 – 2013

LANGUAGES

Vietnamese (native), English (fluent)